

Polya, Problem Solving, and Education
Author(s): Alan H. Schoenfeld
Source: *Mathematics Magazine*, Vol. 60, No. 5 (Dec., 1987), pp. 283-291
Published by: Mathematical Association of America
Stable URL: http://www.jstor.org/stable/2690409
Accessed: 07/02/2010 12:48

Your use of the JSTOR archive indicates your acceptance of JSTOR's Terms and Conditions of Use, available at http://www.jstor.org/page/info/about/policies/terms.jsp. JSTOR's Terms and Conditions of Use provides, in part, that unless you have obtained prior permission, you may not download an entire issue of a journal or multiple copies of articles, and you may use content in the JSTOR archive only for your personal, non-commercial use.

Please contact the publisher regarding any further use of this work. Publisher contact information may be obtained at http://www.jstor.org/action/showPublisher?publisherCode=maa.

Each copy of any part of a JSTOR transmission must contain the same copyright notice that appears on the screen or printed page of such transmission.

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.



Mathematical Association of America is collaborating with JSTOR to digitize, preserve and extend access to *Mathematics Magazine.*

http://www.jstor.org

Pólya, Problem Solving, and Education

ALAN H. SCHOENFELD
University of California
Berkeley, CA 94720

Nineteen forty-five was both a banner year and a year of great chaos for problem solving. It was the year that Max Wertheimer's *Productive Thinking*, a classic study of problem solving, first appeared in English. Jacques Hadamard's "Essay on the Psychology of Invention in the Mathematical Field" appeared as well. So did Karl Duncker's monograph *On Problem Solving*. At the same time mathematics instruction in the schools was mostly drill-and-practice, since it was based on the Associationists' "theory of bonds" (see [**16**]). Drill-and-practice was under attack, however. With Wertheimer leading the charge, the Gestaltists claimed that such instruction completely missed the substance of mathematical thinking. The Gestaltists argued that rote instruction was of little value, and that when students memorized without understanding they missed the underlying essence of the mathematics they studied. Unfortunately, the Gestaltists had no theory of learning or instruction, since they believed all the real "action" takes place in the subconscious. Hence they had few practical classroom suggestions. Moreover, they were counterattacked by the behaviorists, who claimed that "mind" and "thinking" were useless constructs, and that all behavior (mathematical and otherwise) could be explained by stimulus-response chains.

It was against that background of psychological and pedagogical confusion that the most important problem-solving work of the time appeared. As indicated in the next section, *How to Solve It* was hardly Pólya's first foray into the world of problem solving. It was, however, an absolutely critical one. *How to Solve It* marked a turning point both for its author and for problem solving. For Pólya it was the first of a series of major volumes on the nature of mathematical thinking, the topic that became the focus of his work in his later years. *How to Solve It* was followed in 1954 by the two volumes of *Mathematics and Plausible Reasoning*, and in 1962 and 1965 by the two volumes of *Mathematical Discovery*. For mathematics education and for the world of problem solving it marked a line of demarcation between two eras, problem solving before and after Pólya. Since then Pólya's influence both on the study of mathematical thinking and on the study of productive thinking in general has been enormous. One major purpose of this note is to trace out the main ideas in Pólya's work. Another is to trace the influence of those ideas on subsequent research into the nature of mathematical thinking.

Pólya and "modern heuristic"

Pólya's interests in the nature of mathematical thinking and in mathematical pedagogy were apparent early in his career. They had already borne significant fruit by 1925, when Pólya and Szegő produced their landmark problem book, *Aufgaben und Lehrsätze aus der Analysis I*. The English version of the book, *Problems and Theorems in Analysis I*, was translated in 1972 and still has the cachet of the original. That book is a good place to start our description of Pólya's work, for its epigraph reads:

> *What is good education? Giving systematically opportunity to the student to discover things by himself.*

It announces the two main pedagogical themes that Pólya was to pursue for more than half a century. The two themes, woven together through Pólya's works, are order (or systematicity) and

The work described in this note was supported by the National Science Foundation through NSF grant MDR-8550332. NSF grant support does not necessarily imply NSF endorsement of the ideas expressed here.

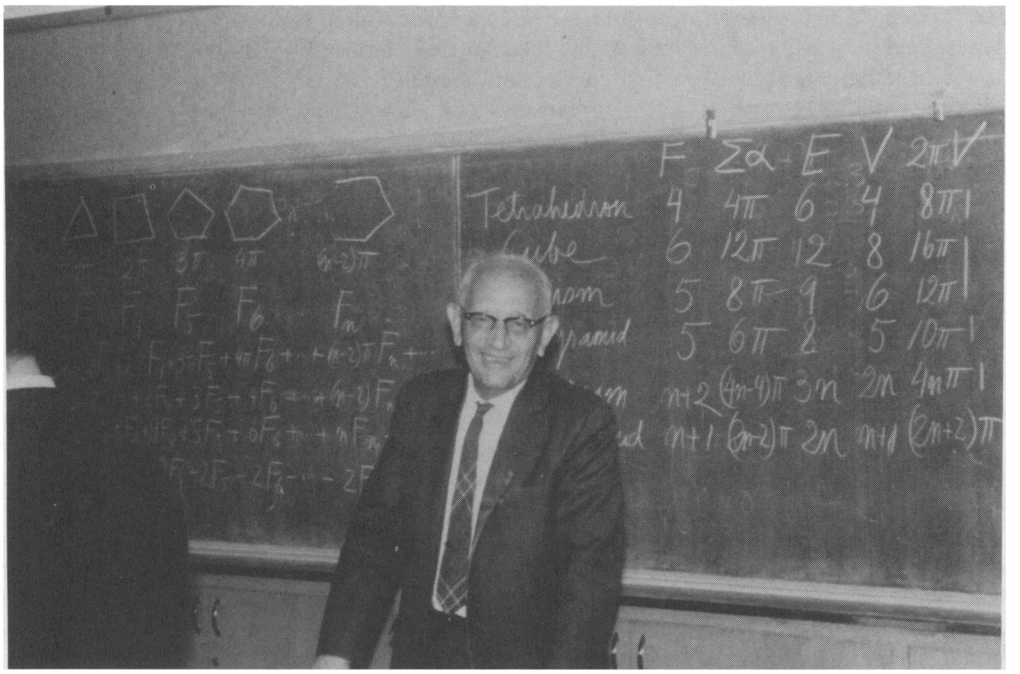
discovery—the systematic discovery of mathematics, and the systematic exposition of a "method" (à la Descartes) for discovery and invention. The two volumes of *Problems and Theorems in Analysis* are primarily concerned with the former. In them Pólya and Szegő seek to reveal the structure of specific mathematical areas through carefully sequenced problems—or as they put it, "to accustom advanced students of mathematics, through systematically arranged problems in some important fields of analysis, to the ways and means of independent thought and research" [**8**, Preface, p. vi]. To this day, anyone who has worked through the problem books has mastered the basics of advanced mathematics, has done some significant problem solving, and is probably ready for serious research. Yet it should be noted that, even then, mathematical structure was not all that concerned Pólya. Although the two volumes of problems are primarily pieces of mathematical exposition, it is clear that when writing them Pólya had "method" on his mind. In the preface to the first (1925) German edition the authors offered a few aphorisms for productive thinking, and the following comments about their use:

> General rules which could prescribe in detail the most useful discipline of thought are not known to us. Even if such rules could be formulated, they could not be very useful... [for] one must have them assimilated into one's flesh and blood and ready for instant use... The independent solving of challenging problems will aid the reader far more than the aphorisms which follow, although as a start these can do him no harm. [**8**, Preface, p. vii.]

Two decades later Pólya returned, at book length, to the same issue. He had embarked on the study of the history of mathematical invention, a pursuit that would come to fruition in the two volumes of *Mathematics and Plausible Reasoning*. Pólya was no longer content with mere aphorisms. He had begun, in the spirit and tradition of Descartes' *Rules for the Direction of the Mind*, an introspective study of *method*—a study of the patterns of productive thinking that enabled him (and would, it was assumed, enable others) to be successful at mathematical problem solving. Perhaps chastened by the failure of Descartes' *Rules* and informed by his historical studies, Pólya took a more modest approach. His was the study of *heuristic*. "The aim of heuristic is to study the methods and rules of discovery and invention... Heuristic, as an adjective, means 'serving to discover.' Heuristic reasoning is reasoning not regarded as final and strict but as provisional and plausible only, whose purpose is to discover the solution of the present problem" [**9**, p. 112–113].

How to Solve It provides the general outline of a problem solving framework and a hint of the details necessary to implement it. The general framework offers a four-phase description of the problem-solving process: understanding the problem, devising a plan, carrying out the plan, looking back. The details are Pólya's heuristic suggestions, or as he calls them, his modern heuristic. Heuristic strategies are rules of thumb for making progress on difficult problems. There are, for example, heuristic strategies for understanding a problem (focusing on the unknown, on the data, drawing a diagram, etc.), for devising a plan (exploiting related problems, analogous problems, working backwards, etc.), and for carrying out and checking a solution. The "short dictionary of heuristic," an alphabetical compendium of techniques and historical notes comprising the bulk of *How to Solve It*, provides some elaboration.

Like the Gestaltists, Pólya was deeply concerned with mathematical structure and used introspection as a primary vehicle for insight. Unlike the Gestaltists, however, Pólya probed deeply into the sources of his insight. As an example illustrating the contrast one may consider Duncker's classic discussion of the *thirteen problem*: "Why are all six-digit numbers of the form *abc*, *abc* divisible by 13?" Gestaltist to the core, Duncker argued that the difficulty with the problem disappears when the fact that all such numbers are divisible by 1001 "emerges from the subconscious." Unfortunately, how or why that particular fact might emerge from the subconscious was left unaddressed. In contrast, Pólya's "modern heuristic" was intended as the vehicle for such insights. How could we deal with the thirteen problem, Pólya would ask, so that we might reasonably expect to discover the factor of 1001? Here are three possible ways, exemplifying three of Pólya's heuristic strategies.



Pólya, c. 1965

Explore the conditions of the problem.
We are given the number *abc*, *abc*. What is this number, really? It's a base 10 shorthand for

$$\begin{aligned} &10^5a + 10^4b + 10^3c + 10^2a + 10b + c \\ &\quad = 100100a + 10010b + 1001c \\ &\quad = 1001(100a + 10b + c), \end{aligned}$$

which reveals the hitherto "mysterious" factor of 1001.

Explore a specific case.
Pick a random three-digit number *abc*, and write it down twice. Say we pick 294, to produce 294,294. What happens when we say this number out loud?

294 thousand and 294

suggests

$$294(1000) + 294 = 294(1001)$$

and we're done.

Look at simple cases and try to find a pattern.
In class, one of my students suggested that we try the simplest 3-digit number, $abc = 001$. This yielded

$$abc, abc = 001001.$$

Her next choice as $abc = 002$, yielding 2002; then 3003, and so on. It was then obvious that each number of the form *abc*, *abc* had a factor of 1001.

Such were the ideas explored, in preliminary fashion, in *How to Solve It*. They were explored at much greater length and depth in the two volumes of *Mathematics and Plausible Reasoning*: *Induction and Analogy in Mathematics* (Vol. 1) and *Patterns of Plausible Inference* (Vol. 2). In

those volumes, filled to the brim with historical examples of mathematical insight and invention, Pólya staked his major philosophical and pedagogical claims. We find them summed up as follows: "Mathematical facts are first guessed and then proved, and almost every passage in this book endeavors to show that such is the normal procedure. If the learning of mathematics has anything to do with the discovery of mathematics, the student must be given some opportunity to do problems in which he first guesses and then proves some mathematical fact on an appropriate level." [**10**, p. 160]

On the philosophical plane Pólya was formulating an empiricist view of mathematics, one that treats mathematics as a discipline of discovery on a par with the natural sciences. Despite the fact that it may appear so from the outside, Pólya argued, mathematics is not a purely deductive science that starts with a few axioms and builds up irrevocably from those. "To a philosopher with a somewhat open mind all intelligent acquisition of knowledge should appear sometimes as a guessing game... The result of the mathematician's creative work is demonstrative reasoning, a proof, but the proof is discovered by plausible reasoning, by guessing." [**10**, p. 158] This claim was backed up by means of extensive historical exegeses which included discussions of Euler's heuristic summation of the series $\Sigma\ 1/n^2$; Newton's extrapolations from projectile motion to the orbits of planets; induction in solid geometry, including some of the development of Euler's formula, $V - E + F = 2$ for polyhedra; developments in number theory, physical mathematics, and the theory of probability. Though our topic here is problem solving and not epistemology, it should be noted that these explorations ultimately had philosophical significance as well. Pólya's mathematical views paralleled the scientific views of Karl Popper, who argued that the development of scientific knowledge consists of much more than merely piling up of mountains of objective "facts." Imre Lakatos was a joint student of Pólya and Popper. In *Proofs and Refutations*, one of the most charming mathematics and philosophy books ever written, Lakatos pursued the history of the development of Euler's formula. His historical treatment shows that basic definitions (like the definition of "polyhedron") are changed by mathematicians when they need to, in the light of new empirical evidence—for example, when a theorem everybody knows ought to be true turns out to be false. In that way, mathematics responds to "data" just as empirical science does. Definitions are revised by mathematicians in the same way that theories are revised by physical scientists.

But let us return to pedagogy. In addition to the philosophical subject matter, the two volumes of *Mathematics and Plausible Reasoning* offered a range of difficult problems through which readers could come to grips with the topics Pólya discussed. Pólya wrote that the book was intended for "students desiring to develop their own ability, and for readers curious to learn about plausible reasoning and its not so banal relations to mathematics. The interests of the teacher are not neglected, I hope, but they are met rather indirectly than directly. I hope to fill that gap some day." [**10**, p. 160]

His attempt to fill that gap came a decade later, in the two (1962 and 1965) volumes of *Mathematical Discovery*. As Peter Hilton notes in the foreword to the (1980) combined paperback edition of *Mathematical Discovery*, the book is "eminently suitable for the needs of all concerned teachers and interested students at the levels of... high school or early undergraduate mathematics." As with Pólya's earlier books, the text follows the logic of the subject matter, placing the burden of inquiry on the student; the student learns by working sequences of "interesting and worthwhile problems." And as with Pólya's other books, the student who manages to work all the problems has done a fair amount of honest-to-goodness mathematics. *Mathematical Discovery* offers a wealth of problems from straightedge and compass geometric constructions, to algebra, recursion, and applications of superposition. Students who work those problems will have learned some important techniques in a way that gets to the structure of the underlying mathematics and that cuts through the standard topical presentations. The books also offer theoretical descriptions of the problem-solving process, supplemented with exercises so that students "live" the mathematics rather than just reading about it. In short, *Mathematical Discovery* is a classic. It's a required text in my undergraduate problem-solving classes and in my classes for high school

mathematics teachers. It represents the capstone of Pólya's career, returning to the themes he explored throughout that career: the structure of mathematics and the nature of mathematical discovery. The combined volumes on those topics—the two problem books co-authored with Szegő, *How to Solve It*, the two volumes of *Mathematics and Plausible Reasoning*, and the two volumes of *Mathematical Discovery*—comprise a collection of extraordinary scope and power.

Heuristics and problem solving since 1945

This section traces developments in mathematical problem solving since 1945. Since problem solving and education lie in the public arena, there are two dimensions along which to consider those developments: the impact of Pólya's work and ideas in the real world and the evolution of research on problem solving, considered as a field of scientific inquiry.

Indicators of Pólya's influence on the real world are plentiful. For example, the National Council of Teachers of Mathematics recommended in its (1980) *Agenda for Action* that "problem solving be the focus of school mathematics in the 1980's." To help move things along the 1980 NCTM yearbook [**4**] was devoted to *Problem Solving in School Mathematics*; subsequent yearbooks and other NCTM publications followed up on the same theme. To get the problem-solving bandwagon rolling, the 1980 Yearbook featured the four-stage problem-solving model from *How to Solve It* on its inside covers. Moreover, virtually every article in the Yearbook is based on Pólya's ideas. In short, "problem solving" in mathematics education means problem solving à la Pólya. And the problem-solving movement is hardly limited to mathematics education; if anything, it's nearly universal. In the past five years Pólya's work on problem solving has been cited in the *American Political Science Review*, *Annual Review of Psychology*, *Artificial Intelligence*, *Computers and Chemistry*, *Computers and Education*, *Discourse Processes*, *Educational Leadership*, *Higher Education*, *Human Learning*, and many more journals. Pólya's real-world impact is clear.

In contrast, the scientific status of heuristic strategies—Do people really use them? Do they work? Can they be taught?—has been problematic.* On the one hand, such strategies have enormous face validity. My own reaction to them was typical. A practicing mathematician, I tripped across Pólya's writings on mathematical thinking about a dozen years ago. I read through them with increasing wonder. Page after page, Pólya described the problem-solving techniques I had learned to use—but had never been taught!—as I did mathematics. In Pólya's writings I recognized both the ways that I did mathematics and the ways that I thought about it. That recognition was shared by most mathematicians, and by most mathematics educators. Pólya's discussions just felt right.

They weren't, however. Or to put it another way, there was ample empirical evidence to suggest that something was either wrong or missing. Excited by my readings in the early 1970s, I sought out some problem-solving experts, mathematics faculty who coached students for the Putnam exam or for various Olympiads. Their verdict was unanimous and unequivocal: Pólya was of no use for budding young problem-solvers. Students don't learn to solve problems by reading Pólya's books, they said. In their experience, students learned to solve problems by (starting with raw talent and) solving lots of problems. The mathematics education literature was no more encouraging. Largely because Pólya's ideas seemed so right, the math-ed literature was chock full of studies designed to teach problem solving *via* heuristics. Unfortunately, the results —whether in first grade, algebra, calculus, or number theory, to name a few—were all depressingly the same, and served to confirm the statements of the Putnam and Olympiad trainers. Study after study produced "promising" results, where teacher and students alike were happy with the instruction (a typical phenomenon when teachers have a vested interest in a new program), but where there was at best marginal evidence (if any!) of improved problem-solving

*Parts of this section are adapted from Schoenfeld [**13**], where more details may be found. The truly masochistic reader can find full documentation for the ideas sketched here in my *Mathematical Problem Solving* [**12**].

performance. Despite all the enthusiasm for the approach, there was no clear evidence that the students had actually learned more as a result of their heuristic instruction, or that they had learned any general problem-solving skills that transferred to novel situations. Finally, Pólya's ideas were challenged on scientific grounds by researchers in artificial intelligence (AI). In their classic book *Human Problem Solving* [7], Newell and Simon describe the genesis of a computer program called General Problem Solver (GPS). GPS was developed to solve problems in symbolic logic, chess, and "cryptarithmetic" (a puzzle domain similar to cryptograms, but with letters standing for numbers instead of letters). GPS played a decent game of chess, solved cryptarithmetic problems fairly well, and managed to prove almost all of the first 50 theorems in Russell and Whitehead's *Principia Mathematica*—all in all, rather convincing evidence that its problem-solving strategies were pretty solid. Now for researchers in AI, the proof is in the empirical pudding: your theory is right if (and, the hard-nosed AI guys would say, only if) you have a running program that instantiates your theory and solves the problems you claimed it would. By those standards, heuristics à la Pólya fell short. One leading AI researcher put it bluntly: "We tried to write problem-solving programs using Pólya's heuristics, and they failed; we tried other methods, and they succeeded. Thus we suspect the strategies he describes are epiphenomenal rather than real—and even if they are real, they're far less important than the ones we use in our programs."

In sum: As of the mid-to-late 1970s, there was empirical reason, both in AI and mathematics education, to doubt the solidity of the heuristical foundations established by Pólya. In the past decade, things have begun to look brighter for heuristics—partly because some of the "general problem-solving techniques" promoted by early AI researchers didn't pan out as hoped, and partly because AI itself (more generally, cognitive science) provided some of the means by which Pólya's ideas could be refined and implemented. What follows is an outline of some developments since the mid-1970's. For details, see Schoenfeld ([**12**], [**13**]) and Silver [**15**].

At a certain level, Pólya's descriptions of problem-solving strategies were right. If you already knew how to use the strategies, you recognized them in his writings. That's why mathematicians resonated with Pólya's work. But at a finer grain size (that is, subjected to a more detailed analysis), Pólya's problem-solving descriptions didn't contain enough detail for people unfamiliar with the strategies to be able to implement them. Pólya's characterizations were *labels* under which families of related strategies were subsumed. One quick example will give the flavor of the analysis. The basic idea is that when you look closely at any single heuristic "strategy," it explodes into a dozen or more related, but fundamentally different, problem-solving techniques. Consider a typical strategy, "examining special cases":

To understand an unfamiliar problem better, you may wish to exemplify the problem by considering various special cases. This may suggest the direction of, or perhaps the plausibility of, a solution.

Now consider the solution to the following three problems.

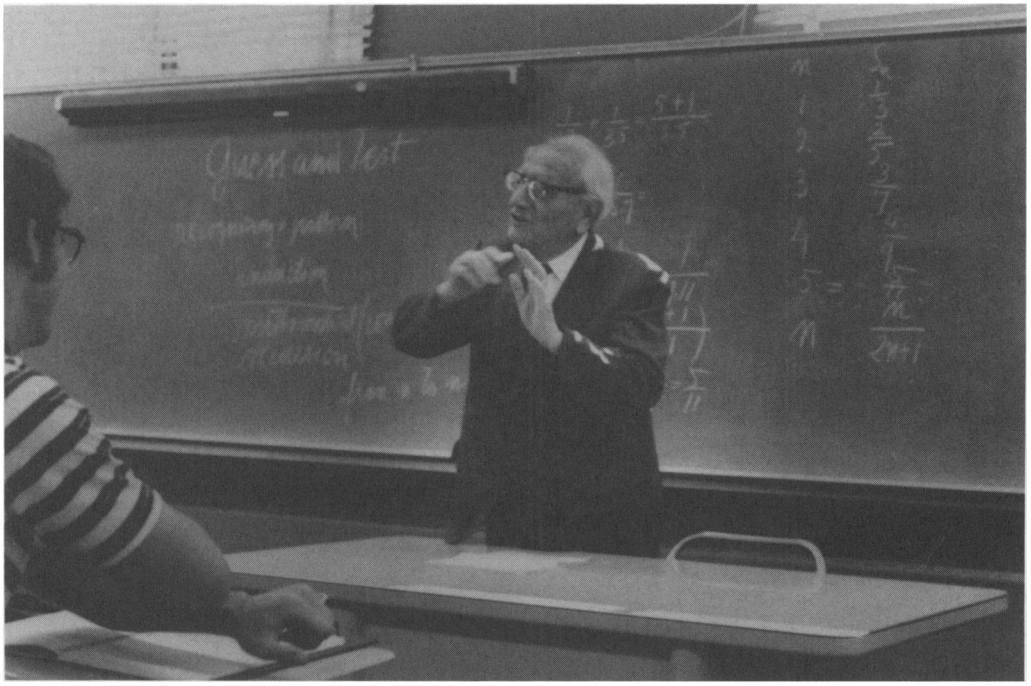
Problem 1. Determine a formula in closed form for the series

$$\sum_{k=1}^{n} k/(k+1)!$$

Problem 2. Let $P(x)$ and $Q(x)$ be polynomials whose coefficients are the same but in "backwards order":

$$P(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n, \quad \text{and}$$
$$Q(x) = a_n + a_{n-1} x + a_{n-2} x^2 + \cdots + a_0 x^n.$$

What is the relationship between the roots of $P(x)$ and $Q(x)$? Prove your answer.



Pólya lecturing at Santa Clara, c. 1978

Problem 3. Let the real numbers a_0 and a_1 be given. Define the sequence $\{a_n\}$ by

$$a_n = 1/2(a_{n-2} + a_{n-1}) \qquad \text{for each } n \geq 2.$$

Does the sequence $\{a_n\}$ converge? If so, to what value?

I'll leave the details of the solutions to you. However, the following observations are important. For problem 1, the special cases that help are examining what happens when the integer parameter, n, takes on the values $1,2,3\ldots$ in sequence; this suggests a general pattern that can be confirmed by induction. Yet if you try to use special cases in the same way on the second problem, you may get into trouble: Looking at values $n = 1,2,3,\ldots$ can lead to a wild goose chase. It turns out that the right special cases of $P(x)$ and $Q(x)$ to look at for problem 2 are easily factorable polynomials. If, for example, you consider

$$P(x) = (2x+1)(x+4)(3x-2),$$

you will discover that its "reverse," Q, is easily factorable. The roots of P and Q are easy to compare, and the result (which is best proved another way) is obvious. And again, the special cases that simplify the third problem are different in nature. If you choose the values $a_0 = 0$ and $a_1 = 1$, you can see what happens for that particular sequence. The pattern in that case suggests what happens in general, and (especially if you draw the right picture!) leads to a solution of the original problem.

Each of these problems typifies a large class of problems, and exemplifies a different special cases strategy. We have:

Strategy 1. When dealing with problems in which an integer parameter n plays a prominent role, it may be of use to examine values of $n = 1,2,3\ldots$ in sequence, in search of a pattern.

Strategy 2. When dealing with problems that concern the roots of polynomials, it may be of use to look at easily factorable polynomials.

Strategy 3. When dealing with problems that concern sequences or series that are constructed recursively, it may be of use to try initial values of 0 and 1—if such choices don't destroy the generality of the processes under investigation.

Needless to say, these three strategies hardly exhaust "special cases." At this level of analysis—the level of analysis necessary for implementing the strategies—one could find a dozen more. This is the case for almost all of Pólya's strategies. In consequence, the two dozen or so "powerful strategies" in *How to Solve It* are, in actuality, a collection of two or three hundred less "powerful," but actually usable, strategies. These strategies *can* be taught—but the fact that there are so many of them causes a new problem. With three hundred techniques potentially at your disposal, you have to know which ones to use, and at what times. "Knowing" the right method won't do you much good, if you don't get around to using it. A brief anecdote illustrates the dilemma.

A number of years ago, I deliberately put the problem

$$\int \frac{x\,dx}{x^2 - 9}$$

as the first problem on a test of techniques of integration, to give my students a boost as they began the exam. With the substitution $u = x^2 - 9$, which I expected the students to use, you can knock the problem off in just a few seconds. Half of the students did this, and got off to a good start. But a fourth of them used the correct but quite time-consuming procedure of partial fractions—and because they spent so much time on the problem, they did poorly on the exam. Half of the rest used the trig substitution $x = 3\sin\theta$—also correct, but so time-consuming that they wound up very far behind and bombed the exam. It is interesting that the students who used the harder techniques showed they knew "more," or at least, more difficult mathematics than the ones who used the easy technique. But they also showed that "it's not just what you know; it's how and when you use it." It's nice when what you do is right, but it's much better when it's also *appropriate*.

The last sentence is too optimistic in tone. The fact is that most of what we do is wrong. That is, most of our guesses—when we're working *real* problems—turn out not to be right. That's natural; the problems wouldn't be problems, but exercises, if that weren't the case. But the hallmark of good problem solvers is that they don't get lost forever in pursuing their wrong guesses. They manage to reject the ones that turn out not to be fruitful, and (eventually!) to hone in on directions that seem to help. Research now indicates that a large part of what comprises competent problem-solving behavior consists of the ability to monitor and assess what one does while working problems, and to make the most of the problem-solving resources at one's disposal. It also indicates that students are pretty poor at this, partly because issues of "resource allocation during thinking" are almost never discussed. But there is evidence that when students get coaching in problem solving that includes attention to such things—when they are encouraged to think about issues like "What are you doing? Why are you doing it? How will it help you solve the problem?"—their problem-solving performance can improve dramatically. (The jargon term for the "resource allocation" aspect of thinking discussed in this paragraph is *metacognition*. For a general introduction to the topic and for more details, the reader might want to look at the chapter entitled "What's all the fuss about metacognition?" in Schoenfeld [**14**].)

To sum up in a sentence, empirical evidence gained over the past decade indicates that Pólya's intuitions may have been right—at least when they're fleshed out with the details made accessible by the tools of cognitive science. So things look better for Pólya on the math-ed front. And on the AI front, things look better as well. In the 1970s researchers in AI were betting on powerful, domain-independent problem-solving strategies such as "means-ends analysis"; domain-specific strategies such as mathematical heuristics were not held in high regard. But the domain-independent strategies have not lived up to their promise, and current cognitive research focuses on the

elaboration of problem-solving strategies tied to bodies of subject matter—e.g., on problem-solving strategies for particular domains of mathematics. We may be close to the point where we can build computer programs employing Pólya-type heuristic strategies. If the research gets to that point—and it may well, in the next decade or two—then Pólya's intuitions about problem solving will have served as part of the foundation for a true "science of thought."

An expanded version of the article, entitled "A brief and biased history of problem solving," appears as a chapter in: F. Curcio, (ed.), Teaching and Learning: A Focus on Problem Solving, National Council of Teachers of Mathematics, Reston, VA, 1987.

References

[1] R. Descartes, Rules for the direction of the mind (E. S. Haldane and G. R. I. Ross, translators), In Great Books of the Western World, Vol. 31, Encyclopedia Brittanica Inc., Chicago, 1952.

[2] K. Duncker, On problem solving, Psychological Monographs, 58, No. 5, American Psychological Association, Washington, D.C.

[3] J. Hadamard, An essay on the psychology of invention in the mathematical field, Princeton University Press, Princeton, 1945.

[4] S. Krulik (ed.), Problem solving in school mathematics, 1980 Yearbook of the National Council of Teachers of Mathematics, National Council of Teachers of Mathematics, Reston, VA, 1980.

[5] I. Lakatos, Proofs and Refutations, Cambridge University Press, Cambridge, 1976.

[6] An Agenda for Action, National Council of Teachers of Mathematics, Reston, VA, 1980.

[7] A. Newell and H. Simon, Human Problem Solving, Prentice-Hall, Englewood Cliffs, NJ, 1972.

[8] G. Pólya and G. Szegő, Aufgaben und Lehrsätze aus der Analysis I, Springer, Berlin, 1925. An English version, Problems and Theorems in Analysis I (D. Aeppli, trans.), was published by Springer, New York, 1972.

[9] G. Pólya, How to Solve It, Princeton University Press, Princeton, 1945.

[10] _____, Mathematics and Plausible Reasoning, 2 vols., Princeton University Press, Princeton, 1954.

[11] _____, Mathematical Discovery, Wiley, New York, vol. 1, 1962, vol. 2, 1965; combined paperback edition, 1981.

[12] A. Schoenfeld, Mathematical Problem Solving, Academic Press, New York, 1985.

[13] _____, Confessions of an accidental theorist, in For the Learning of Mathematics, February, 1987.

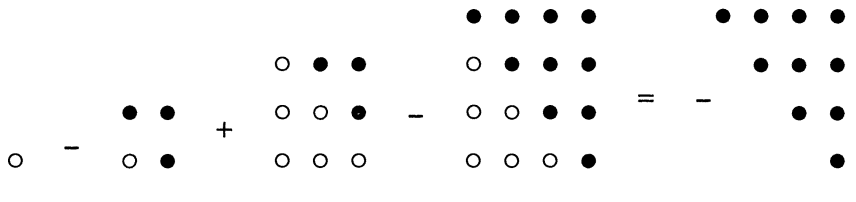
[14] A. Schoenfeld (ed.), Cognitive Science and Mathematics Education, Lawrence Erlbaum, Hillsdale, NJ, 1987.

[15] E. A. Silver (ed.), Teaching and Learning Mathematical Problem Solving: Multiple Research Perspectives, Lawrence Erlbaum, Hillsdale, NJ, 1988.

[16] E. L. Thorndike, The Psychology of Arithmetic, Macmillan, New York, 1922.

[17] M. Wertheimer, Productive Thinking, Harper and Row, New York, 1945/1959.

Proof without Words:



$$\sum_{k=1}^{n} (-1)^{k+1} k^2 = (-1)^{n+1} T_n = (-1)^{n+1} \frac{n(n+1)}{2}.$$

—DAVE LOGOTHETTI
Santa Clara University